DISEN

DISEN ELECTRONICS CO.,LIMITED

■ Preliminary Specification
□ Approval Specification

SPECIFICATION
FOR
LCD MODULE

Customer :

Product Model: DS070HSD39T-132

Sample code :

Designed by	Checked by	Approved by
Sally	Atman	Joune

Final Approval by Customer

□ LCM Machinery OK **Checked By**	□ LCM OK
□ LCM Display OK **Checked By**	□ NG，Problem survey: **Approved By**

※ **The specification of "TBD" should refer to the measured value of sample . If there is difference between the design specification and measured value, we naturally shall negotiate and agree to solution with customer.**

Revision History

Version	Contents	Date	Note
A0	Original	2025.6.10	-

Contents

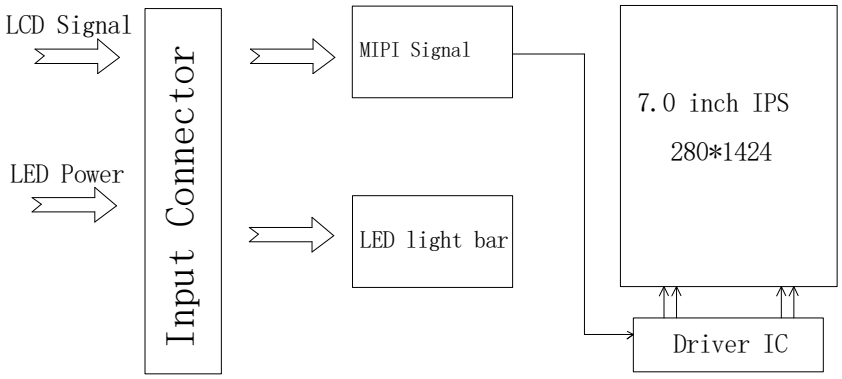
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1. Numbering System

DS □□□ □□□ □□ □ -□□□
1 2 3 4 5 6

No	Definition	Specifications
(1)	TFT LCM Productor No.	DS ---- DISEN ELECTRONICS Co.,Ltd
(2)	Display monitor opposite angle line size	Unit :Inch
(3)	Productor Types	BOE; Innolux ;AUO; HSD; LG ;Panasonic； Tianma;Samsung; IVO; CMI ； Sharp
(4)	Productor Development Series No.	By two figures characters expression from 01 to 99
(5)	With Touch Panel Or Not	T----With T/P ； N----Without T/P
(6)	Productor Development edition No.	By The English letters : 001~ ZZZ

2. Block Diagram

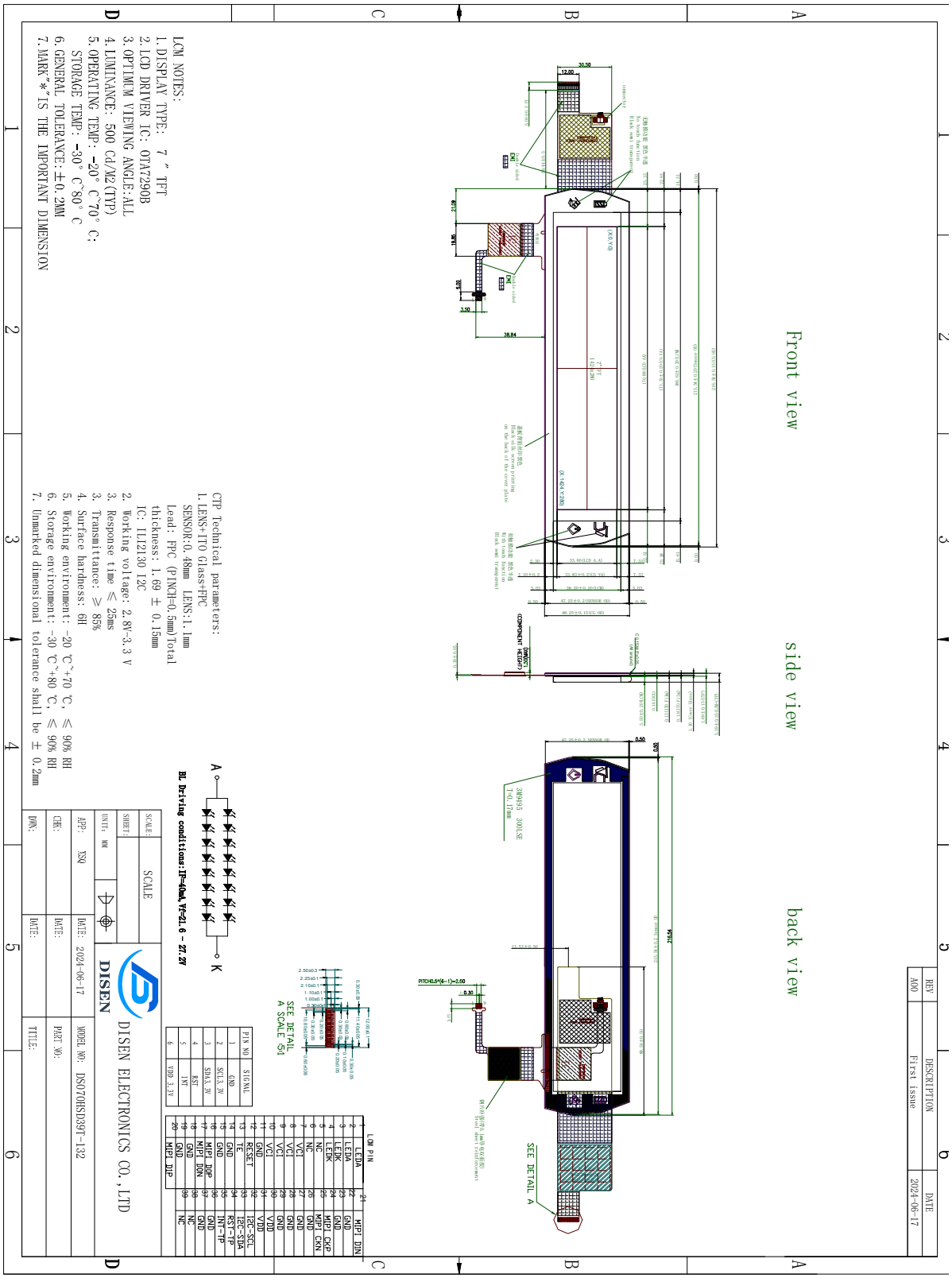


3. Technology Specifications

3.1 LCM General Information

Item	Specification	Unit
LCD Size	7.0	inch
Number of Pixels	280 (H) RGB x 1424 (V)	pixels
Display Mode	Normally Black	-
Viewing Direction	Free	o' clock
Interface	MIPI	-
Display Colors	16.7M	colors
Outline Dimension	48.25 (H) x 216.54 (V) x 5.39 (D)	mm
Active Area	33.60 (H) x 170.88 (V)	mm
Pixel Pitch	0.12 (H) x 0.12 (V)	mm
Driver IC	OTA7290B	-
Operation Temperature	-20~70	℃
Storage Temperature	-30~80	℃

4.Outline Dimension



5.Interface Pin Assignment

Recommended connector：FH26-39S-0.3SHW manufactured by HIROSE

No.	Symbol	Description
1	LEDA	Power for LED backlight (Anode)
2	LEDA	Power for LED backlight (Anode)
3	LEDK	Power for LED backlight (Cathode)
4	LEDK	Power for LED backlight (Cathode)
5-6	NC	No connection
7-10	VCI	Power supply
11	GND	Ground
12	RESET	Global reset pin
13	TE	Tearing effect signal is used to synchronize MCU to frame memory writing，If
14	GND	Ground
15	GND	Ground
16	MIPI-D0P	MIPI Positive data signal(+)
17	MIPI-D0N	MIPI Negative data signal(-)
18	GND	Ground
19	GND	Ground
20	MIPI-D1P	MIPI Positive data signal(+)
21	MIPI-D1N	MIPI Negative data signal(-)
22	GND	Ground
23	GND	Ground
24	MIPI-CKP	MIPI Positive clock signal(+)
25	MIPI-CKN	MIPI Negative clock signal(-)
26-29	GND	Ground
30	VDD	Power supply
31	VDD	Power supply
32	I2C-SCL	I2C clock input
33	I2C-SDA	I2C data input and output
34	RST-TP	Reset pin
35	INT-TP	Interrupt signal from CTP
36	GND	Ground
37	GND	Ground
38	NC	No connection
39	NC	No connection

6 .Operation Specifications

6.1 Absolute Max. Rating

Item	Symbol	MIN.	MAX.	Unit	Note
Analog Supply voltage	VDD	-0.3	5.0	V	Note 1

Note 1:Permanent damage may occur to the LCD module if beyond this specification. Functional operation should be restricted to the conditions described under normal operating conditions.

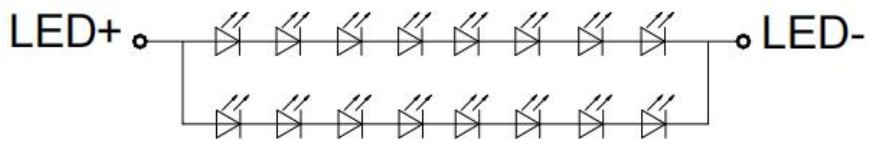
6.2 Operating Conditions

Item	Symbol	Min.	Typ.	Max.	Unit	Note
Analog Supply voltage	VCI	3.0	3.3	3.6	V	
Analog supply current	I_{VCI}	-	TBD	-	mA	VCI=3.3V
Logic input voltage	V_{IH}	0.7*VCI	-	VCI	V	
	V_{IL}	GND	-	0.3*VCI	V	

6.3 Current for LED Driver

Item	Symbol	Min.	Typ.	Max.	Unit	Note
Driving Current	IF	-	40	-	mA	
Driving Voltage	VF	21.6	-	27.2	V	
Power consumption	W_{BL}	0.864	-	1.088	W	
LED Life-Time	N/A	-	50,000	-	Hours	Ta=25℃ Note 1

Note 1:LED lifetime is defined as the module brightness decay 50% of original brightness at Ta=25 degree, typical current.



6.4 Touch Panel

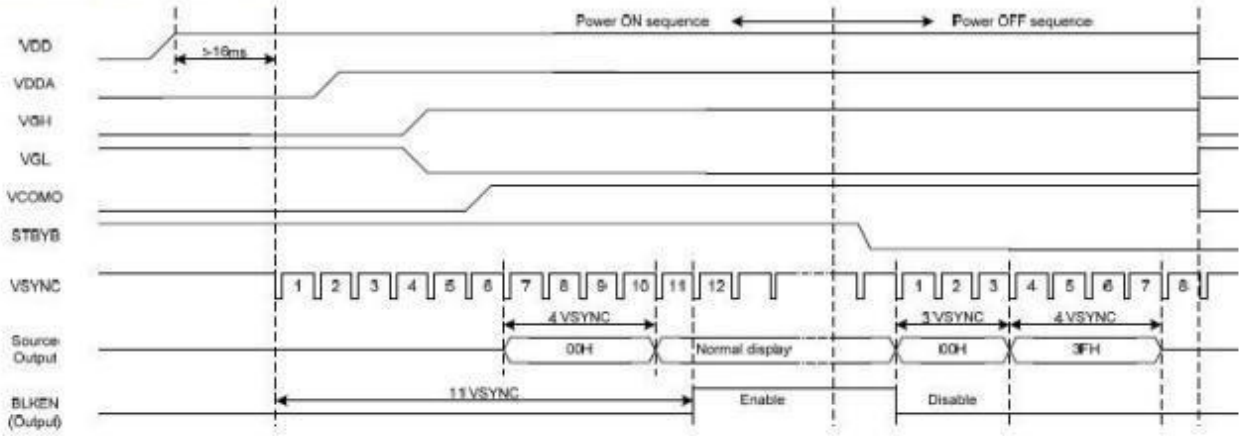
Item	Symbol	Min.	Typ.	Max.	Unit	Note
Power Supply voltage	VDD	-	3.3	-	V	
Analog supply current	I_{VDD}	-	TBD	-	mA	VDD=3.3V
Input high-level voltage	V_{IH}	0.7*VDD	-	VDD	V	
Input low -level voltage	V_{IL}	GND	-	0.3*VDD	V	

7.Signal Timing Characteristics

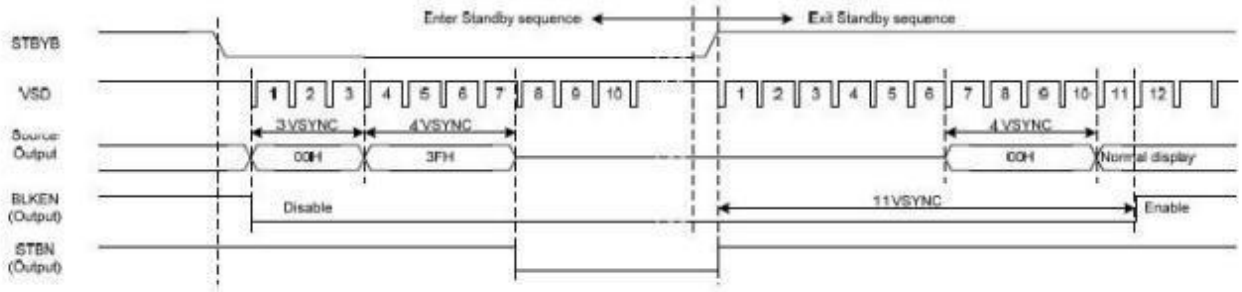
7.1 Power On/Off Sequence

In order to prevent IC from power on reset fail, the rising time (T_{POR}) of the digital power supply VDD should be maintained within the given specifications. Refer to "AC Characteristics" for more detail on timing.

Power-On/Off Timing Sequence:

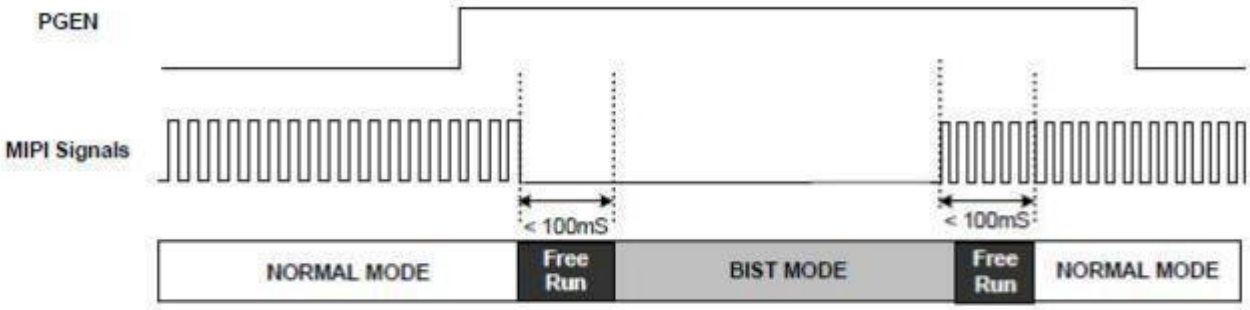


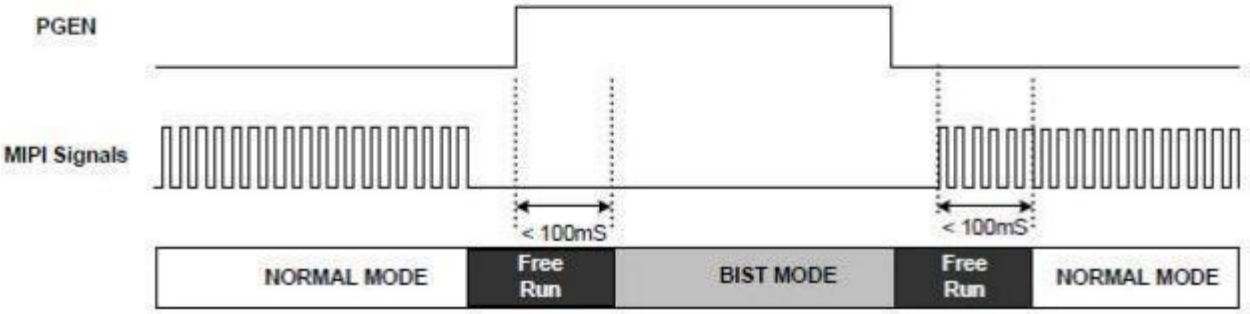
Enter and Exit Standby Mode Sequence:



7.2 Reset Timing Characteristics

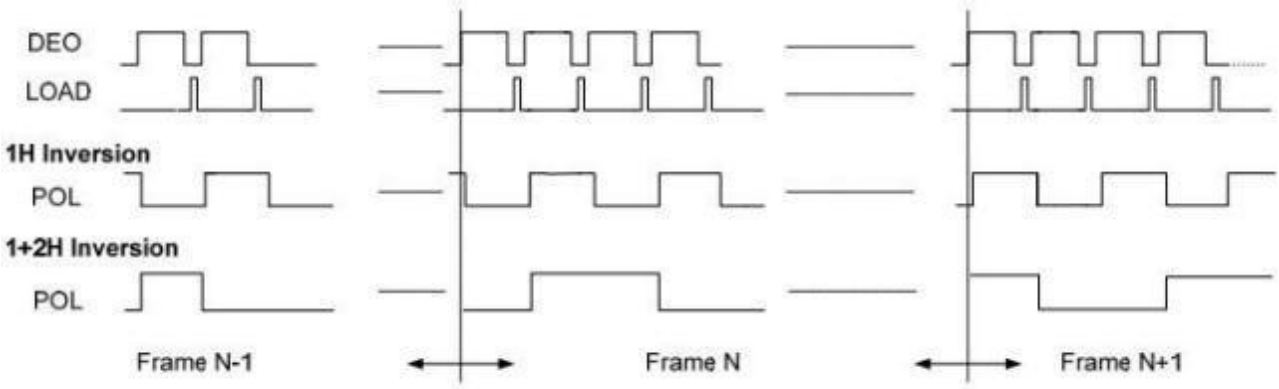
BIST Timing



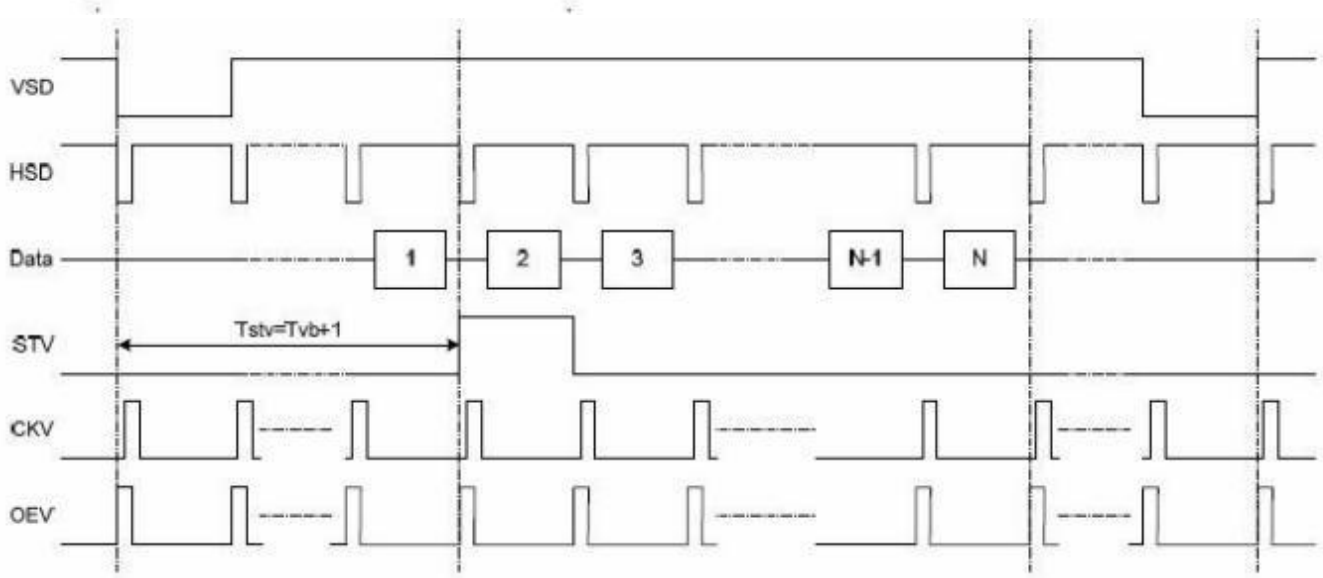


TCON Timing

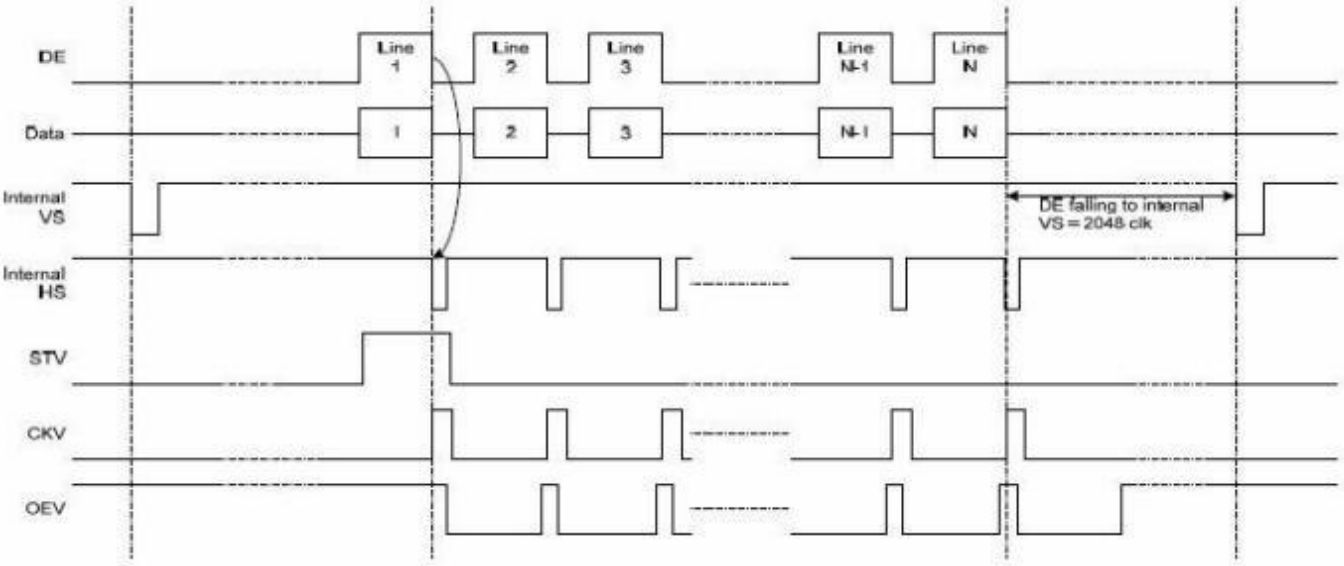
Polarity Control



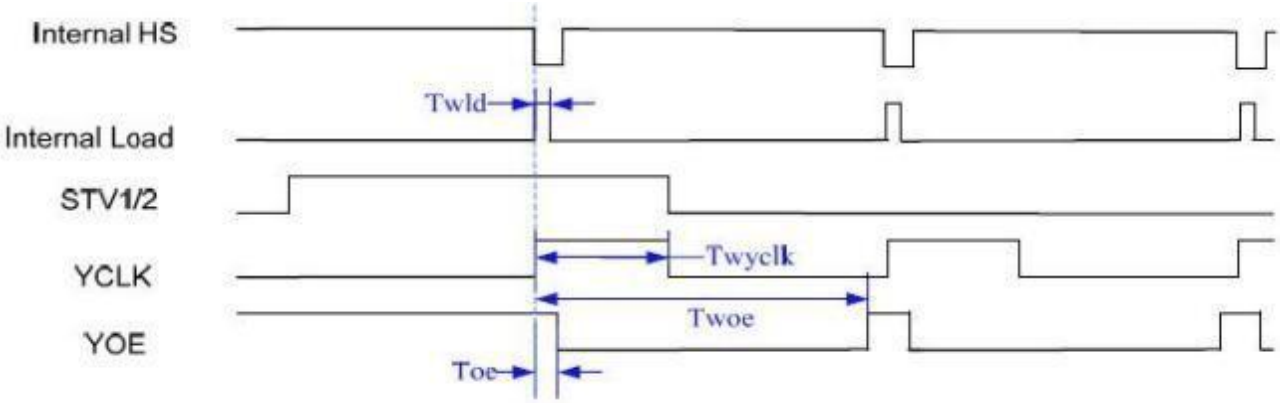
Vertical Timing Diagram HV (Cascade)



Vertical Timing Diagram DE (Cascade)



Gate output timing diagram (Cascade)



MIPI DC Characteristics

HS Receiver DC Specification

Parameter	Symbol	Rating			Unit	Note
		Min	Typ	Max		
Operation Voltage	VDD	1.5-10%	1.5	1.5+10%	mV	
Differential Input Voltage	|VID|	70	200	260	mV	
Common Mode Voltage	$V_{CMRX(DC)}$	70	-	330	mV	
Differential Input High Threshold Voltage	VTH	-	-	70	mV	
Differential Input Low Threshold Voltage	VTL	-70	-	-	mV	
Singled-ended input high voltage	V_{HHS}	-	-	460	mV	
Singled-ended input low voltage	V_{LHS}	-40	-	-	mV	
Singled-ended threshold for HS termination enable	$V_{TERM-EN}$	-	-	450	mV	
Differential input impedance	Z_{ID}	80	100	125	ohm	
Pin leakage current	I_{LEAK}	-10	-	10	uA	
Common-mode interference beyond 450MHz	$\Delta V_{CMRX(HF)}$	-	-	100	mV	
Common-mode interference 50MHz - 450MHz	$\Delta V_{CMRX(LF)}$	-50	-	50	mV	
Common-mode termination	C_{CM}	-	-	60	pF	
Embedded Termination	R_T	90	100	110	ohm	2bits RT_SEL[1: 0] for termination resistor selection 00 → 200ohm 10 , 01→ 150ohm 11 → 100ohm (default) 1bit ERMR_EN for termination resistor enable TERMR_EN=0, termr disable R=(OPEN) TERMR_EN=1, termr enable

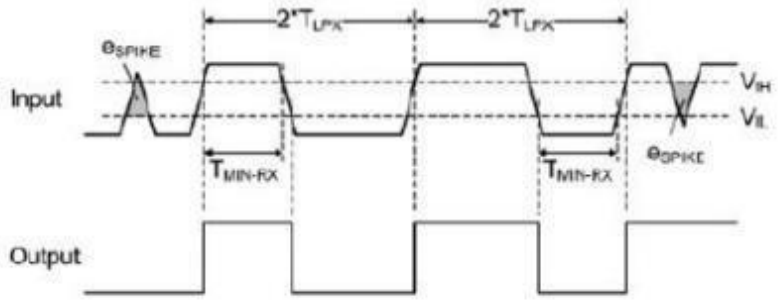
Note:

(1) Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz.

(2) This table value includes a ground difference of 50mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450MHz.

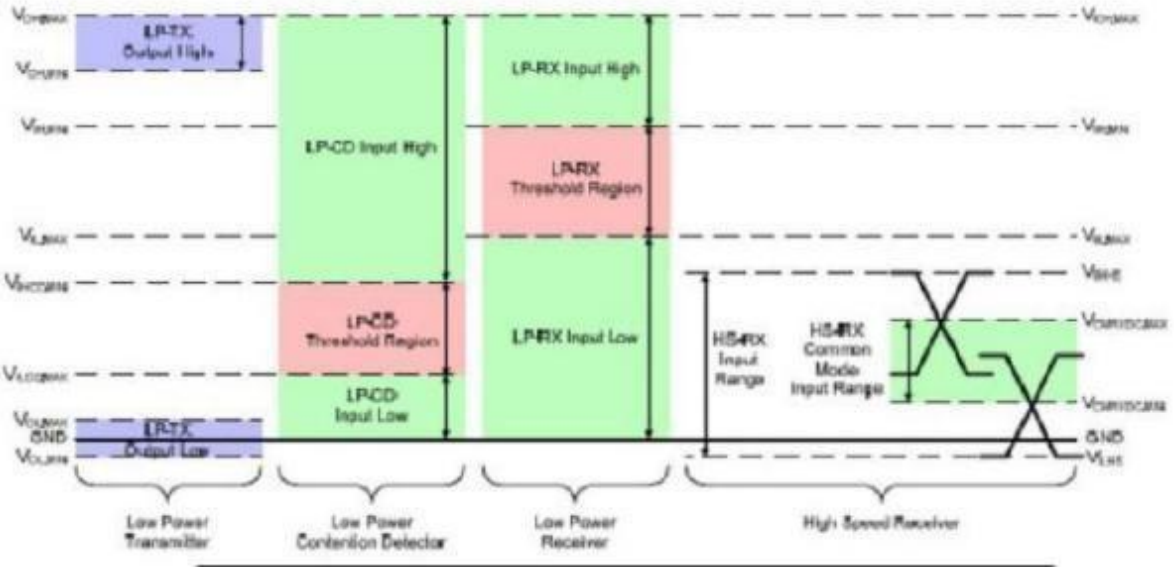
LP Receiver DC Specification

Parameter	Symbol	Rating			Unit	Note
		Min	Typ	Max		
Logic 1 input voltage	V_{IH}	880	-	-	mV	
Logic 0 input voltage, not in ULP State	V_{IL}	-	-	550	mV	
Input hysteresis	V_{HYST}	25	-	-	mV	



Line Contention Detection

Parameter	Symbol	Rating			Unit	Note
		Min	Typ	Max		
Logic 1 contention threshold	V_{IHCD}	450	-	-	mV	
Logic 0 contention threshold	VILCD	-	-	200	mV	



MIPI AC Chrematistics

HS Receiver AC Timing Characteristics

Parameter	Symbol	Rating			Unit	Note
		Min	Typ	Max		
Bandwidth per lane	-	-	-	1000	Mbps	Bandwidth selected by register 'speedup' Speedup=0 → Max=550Mbps Speedup=1→ Max=1000Mbps
Operation frequency	-	-	-	500	MHz	
UI instantaneous	UI_{INST}	1	-	12.5	ns	1
Data to Clock Skew	Tskew	-0.15	-	0.15	UI_{INST}	
Inter-lane static skew	Tskew-lane	-	-	$UI_{INST}/50$	UI_{INST}	
Data to Clock Setup Time	T_{SETUP}	0.25	-	-	UI_{INST}	2
Data to Clock Hold Time	T_{HOLD}	0.25	-	-	UI_{INST}	
Common-mode interference beyond 450MHz	$\Delta V_{CMRX(HF)}$	-	-	100	mV	4
Common-mode interference 50MHz- 450MHz	$\Delta V_{CMRX(LF)}$	-50	-	50	mV	3,6
Common-mode termination	C_{CM}	-	-	60	pF	5

Note:

(1) Total silicon and package delay budget of 0.3*UINST

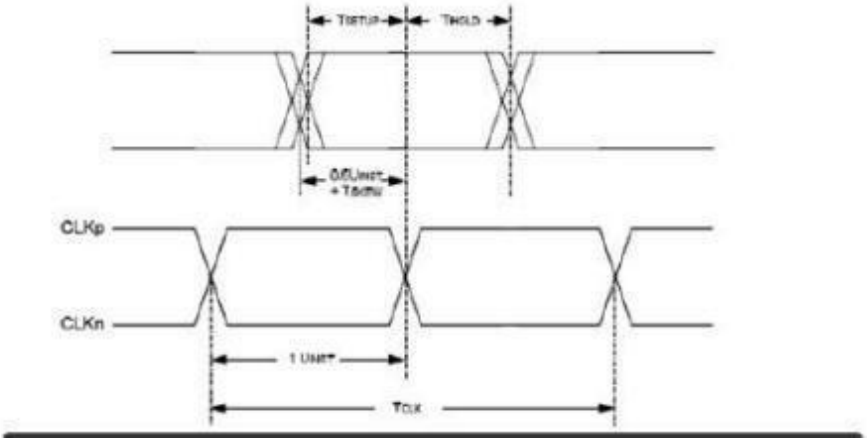
(2) Total setup and hold window for receiver of 0.3*UINST

(3) Excluding 'static' ground shift of 50mV

(4) =VCMRX (HF) is the peak amplitude of a sine wave superimposed on the receiver input

(5) For higher bit rates a 14pF capacitor will be needed to meet the common-mode return loss specification.

(6) Voltage difference compared to the DC average common-mode potential.



8.Optical specifications

Item	Symbol	Condition	Values			Unit	Remark
			Min.	Typ.	Max.		
Viewing angle (CR≥ 10)	θ_L	Φ=180° (9 o'clock)	75	80	-	degree	Note 1
	θ_R	Φ=0°(3 o'clock)	75	80	-		
	θ_T	Φ=90° (12 o'clock)	75	80	-		
	θ_B	Φ=270° (6 o'clock)	75	80	-		
Response time Rise+ Fall	T_{RT}	Normal θ=Φ=0°	-	30	40	msec	Note 3
Contrast ratio	CR		800	1000	-	-	Note 4
Color chromaticity	W_X		Typ. -0.03	0.319	Typ. +0.03	-	Note 2 Note 5 Note 6
	W_Y			0.343		-	
Luminance	L		400	500	-	-	Note 6
Luminance uniformity	Y_U		75	80	-	%	Note 6,7
Color Gamut	NTSC	CIE1931	-	60	-	%	

Note 1: Definition of viewing angle range

Normal line
θ=Φ=0°

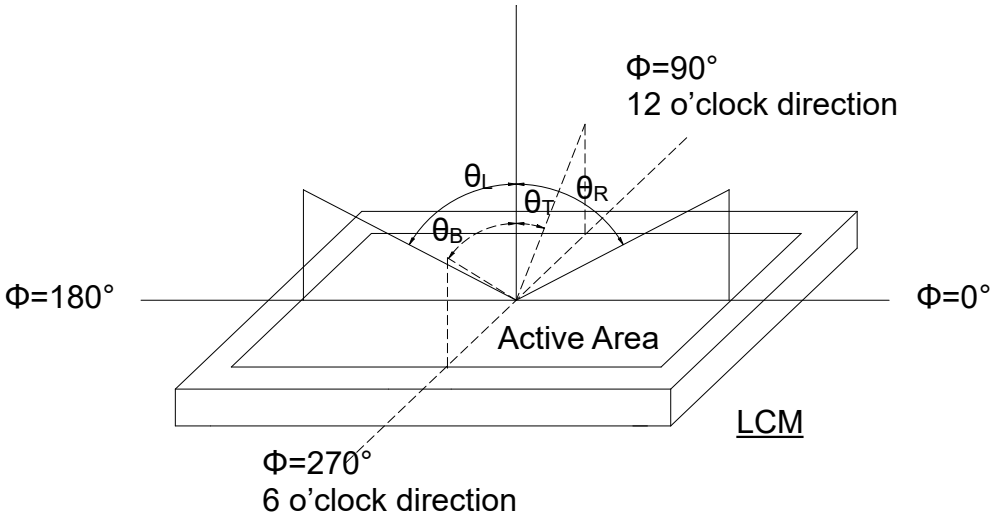


Fig. 4-1 Definition of viewing angle

Note 2: Definition of optical measurement system.

The optical characteristics should be measured in dark room. After 30 minutes

operation, the optical properties are measured at the center point of the LCD screen. (Viewing angle is measured by ELDIM-EZ contrast/Height :1.2mm ,Response time is measured by Photo detector TOPCON BM-5A, other items are measured by BM-7A/Field of view: 1° /Height: 500mm.)

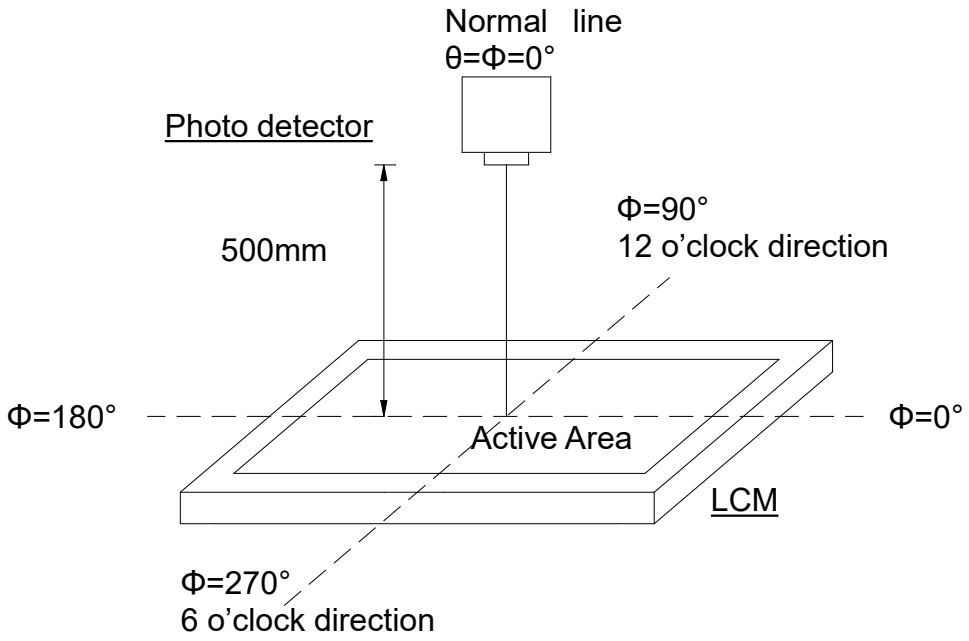


Fig. 4-2 Optical measurement system setup

Note 3: Definition of Response time

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (T_{ON}) is the time between photo detector output intensity changed from 90% to 10%. And fall time (T_{OFF}) is the time between photo detector output intensity changed from 10% to 90%.

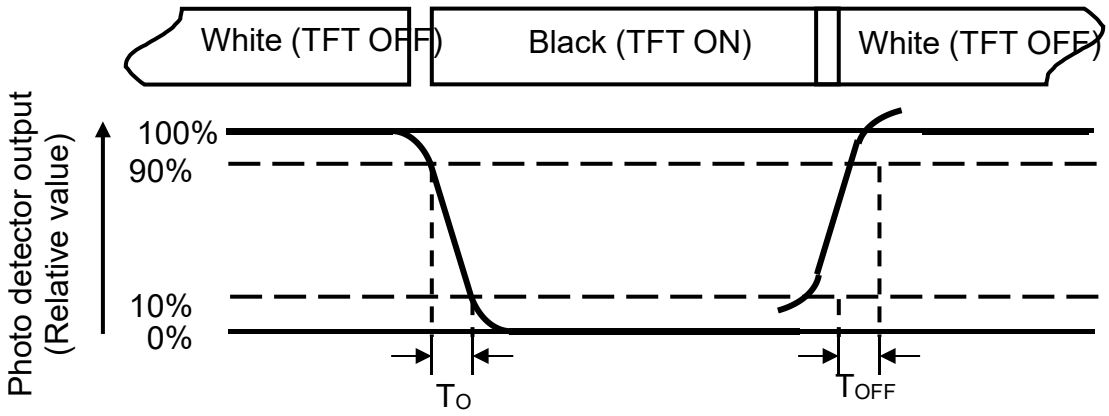


Fig. 4-3 Definition of response time

Note 4: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD on the "White" state}}{\text{Luminance measured when LCD on the "Black" state}}$$

Note 5: Definition of color chromaticity (CIE1931)
Color coordinates measured at center point of LCD.
Note 6: All input terminals LCD panel must be ground while measuring the center area of the panel.

Note 7: Definition of Luminance Uniformity
Active area is divided into 9 measuring areas (Refer to Fig. 4-4).Every measuring point is placed at the center of each measuring area.

$$Luminance\ \ Uniformity\ (Yu) = \frac{B_{min}}{B_{max}}$$

W-------Active area length H----- Active area width

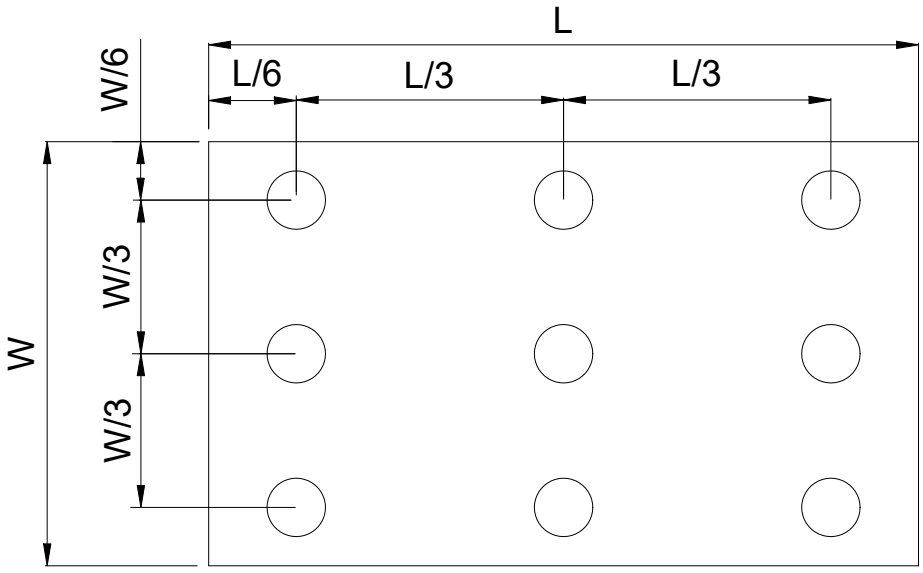


Fig. 4-4 Definition of measuring points

B_{max}: The measured maximum luminance of all measurement position.

B_{min}: The measured minimum luminance of all measurement position.

Note 8: Definition of Color of CIE Coordinate and NTSC Ratio.

$$S = \frac{\text{area of RGB triangle}}{\text{area of NTSC triangle}} \times 100\%$$

9. Reliability Test Conditions And Methods

Test Item	Test Conditions	
High Temperature Storage	Ta= +80℃	96hrs
Low Temperature Storage	Ta= -30℃	96hrs
High Temperature Operation	Ta= +70℃	96hrs
Low Temperature Operation	Ta= -20℃	96hrs
High Temperature and Humidity Storage	Ta= +60℃, 90% RH	96hrs
Thermal Shock (Non-operation)	-30℃/30 min ~ +80℃/30 min for 20 cycles Start with cold temperature end with high temperature	
Electro Static Discharge	Contact = ± 4 kV, class B Air = ± 8 kV, class B R=330Ω,C=150pF	
Vibration	Sweep: 10Hz~55Hz~10Hz Stroke: 1.5mm 2 hrs for each direction of X .Y. Z.	
Mechanical Shock	60G 6ms,±X,±Y,±Z 3 times for each direction	
Package Drop Test	Height: 60 cm 1 corner, 3 edges, 6 surfaces	

Notes: The test result shall be evaluated after the sample has been left at room temperature and humidity for
2 hours without load. No condensation shall be accepted. The sample will not be accepted if appear these defects:

1). Air bubble in the LCD
2). Seal leak or Glass crack
3). Non display or abnormal display
4). Brightness reduction >50%

10. Handling Precautions

10.1 Mounting method

The LCD panel of DISEN LCD module consists of two thin glass plates with polarizes which easily be damaged. And since the module in so constructed as to be fixed by utilizing fitting holes in the printed circuit board.

Extreme care should be needed when handling the LCD modules.

10.2 Caution of LCD handling and cleaning

When cleaning the display surface, Use soft cloth with solvent [recommended below] and wipe lightly

- Isopropyl alcohol
- Ethyl alcohol

Do not wipe the display surface with dry or hard materials that will damage the polarizer surface.

Do not use the following solvent:

- Water
- Aromatics

Do not wipe ITO pad area with the dry or hard materials that will damage the ITO patterns

Do not use the following solvent on the pad or prevent it from being contaminated:

- Soldering flux
- Chlorine (Cl) , Salfur (S)

If goods were sent without being silicon coated on the pad, ITO patterns could be damaged due to the corrosion as time goes on.

If ITO corrosion happen by miss-handling or using some materials such as Chlorine (Cl), Salfur (S) from customer, Responsibility is on customer.

10.3 Caution against static charge

The LCD module use C-MOS LSI drivers, so we recommended that you:

Connect any unused input terminal to Vdd or Vss, do not input any signals before power is turned on, and ground your body, work/assembly areas, assembly equipment to protect against static electricity.

10.4 packing

- Module employ LCD elements and must be treated as such.
- Avoid intense shock and falls from a height.
- To prevent modules from degradation, do not operate or store them exposed direct to sunshine or high temperature/humidity

10.5 Caution for operation

- It is an indispensable condition to drive LCD's within the specified voltage limit since the higher voltage then the limit cause the shorter LCD life.
- An electrochemical reaction due to direct current causes LCD's undesirable deterioration, so that the use of direct current drive should be avoided.
- Response time will be extremely delayed at lower temperature then the operating temperature range and on the other hand at higher temperature LCD's how dark color in them. However those phenomena do not mean malfunction or out of order with LCD's, which will come back in the specified operation temperature.
- If the display area is pushed hard during operation, some font will be abnormally

displayed but it resumes normal condition after turning off once.

- A slight dew depositing on terminals is a cause for electro-chemical reaction resulting in terminal open circuit.

Usage under the maximum operating temperature, 50%Rh or less is required.

10.6 storage

In the case of storing for a long period of time for instance, for years for the purpose or replacement use, the following ways are recommended.

- Storage in a polyethylene bag with the opening sealed so as not to enter fresh air outside in it . And with no desiccant.
- Placing in a dark place where neither exposure to direct sunlight nor light's keeping the storage temperature range.
- Storing with no touch on polarizer surface by the anything else.

[It is recommended to store them as they have been contained in the inner container at the time of delivery from us

10.7 Safety

- It is recommendable to crash damaged or unnecessary LCD's into pieces and wash off liquid crystal by either of solvents such as acetone and ethanol, which should be burned up later.
- When any liquid leaked out of a damaged glass cell comes in contact with your hands, please wash it off well with soap and water

11. Precaution for use

11.1

A limit sample should be provided by the both parties on an occasion when the both parties agreed its necessity. Judgment by a limit sample shall take effect after the limit sample has been established and confirmed by the both parties.

11.2

On the following occasions, the handing of problem should be decided through discussion and agreement between responsible of the both parties.

- When a question is arisen in this specification
- When a new problem is arisen which is not specified in this specifications
- When an inspection specifications change or operating condition change in customer is reported to DISEN , and some problem is arisen in this specification due to the change
- When a new problem is arisen at the customer's operating set for sample evaluation in the customer site.

12. Package Drawing

TBD